

Future Collider Perspectives on Higgs CP Violation

FCC-ee VS LCF from ZH production

Christoph Englert(Glasgow/Manchester), Jay
Nesbitt(Manchester/Glasgow), Andrew
Pilkington(Manchester), Aidan Robson(Glasgow),
Yuyang Zhang(Glasgow/DESY)



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Future Collider Status

Plan A: FCC-ee

Working point	Z pole	WW thresh.	ZH	$t\bar{t}$	
$\sqrt{s}$ (GeV)	88, 91, 94	157, 163	240	340–350	365
Lumi/IP ($10^{34} \text{ cm}^{-2} \text{ s}^{-1}$)	140	20	7.5	1.8	1.4
Lumi/year (ab^{-1})	68	9.6	3.6	0.83	0.67
Run time (year)	4	2	3	1	4
Integrated lumi. (ab^{-1})	205	19.2	10.8	0.42	2.70

Plan B: LCF

	91 GeV	250 GeV	350 GeV	550 GeV	1-3 TeV
$\int \mathcal{L} \text{ (ab}^{-1}\text{)}$	0.1	3	0.4	8	8
beam polarisation (e^-, e^+ ; %)	80/30	80/30	80/30	80/60	80/20
($--, -+, +- , ++$) (%)	(10,40,40,10)	(5,45,45,5)	(5,68,22,5)	(10,40,40,10)	(10,40,40,10)

At 240/250 GeV: Luminosity VS Polarization !



Motivation for searching CPV: Matter-Antimatter Asymmetry

Baryon Asymmetry

$$Y_b = \frac{1}{7.04} \frac{n_b - \bar{n}_b}{n_\gamma} = \begin{cases} (8.2-9.4) \times 10^{-11}, & (\text{BBN}), \\ (8.65 \pm 0.09) \times 10^{-11}, & (\text{CMB}). \end{cases}$$

Dynamic baryogenesis mechanism: The Sakharov Conditions (1967)

- Baryon number violation
- Charge(C) and charge-parity(CP) violation
- Departure from thermal equilibrium

Standard Model CPV source :

CP transformation:

$$\text{CP} |p, s, q\rangle = \eta_{\text{CP}} | -p, s, -q\rangle$$

Left-handed particle

-> Right-handed antiparticle

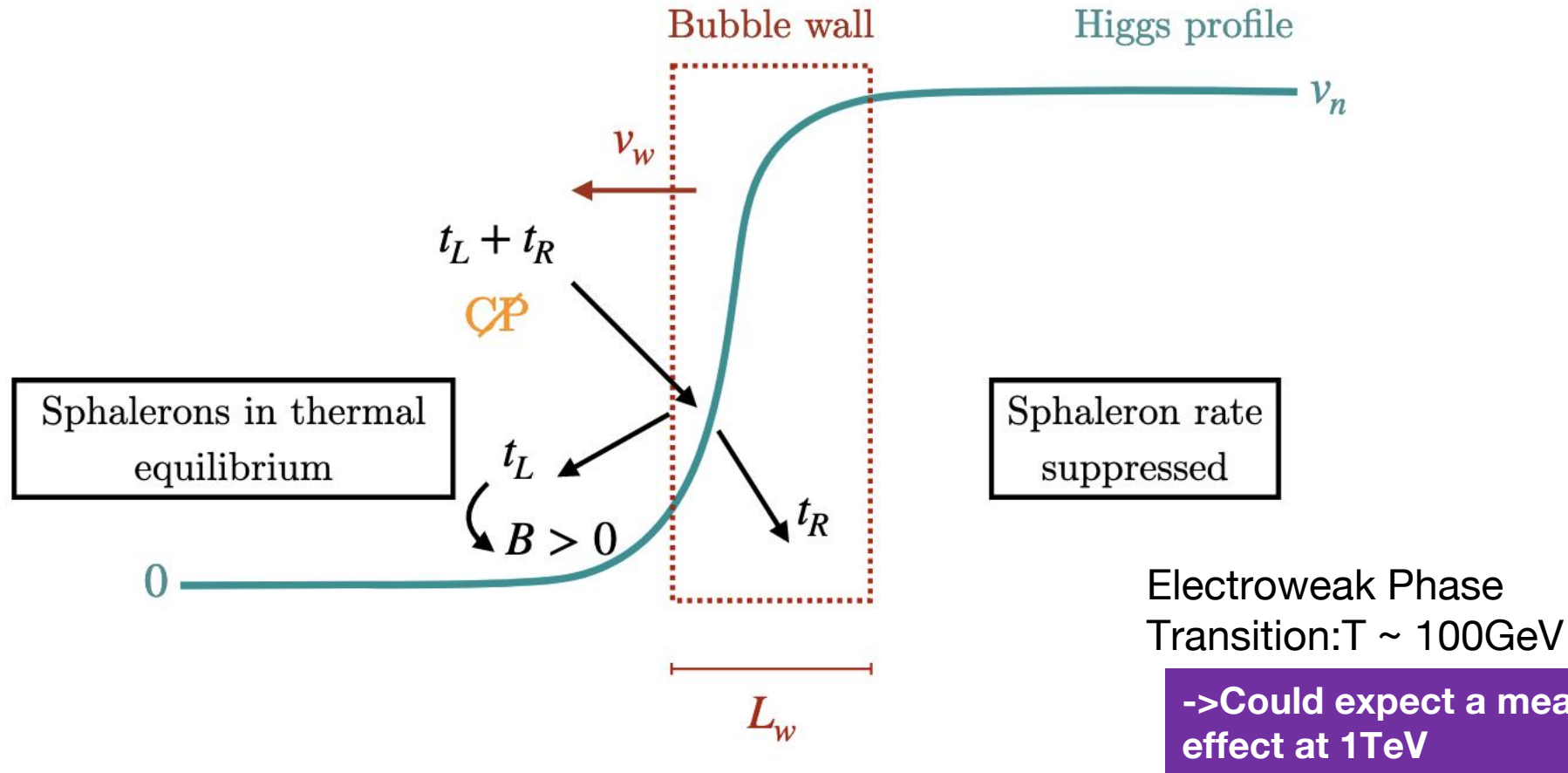
$$\frac{J_{\text{CP}}}{(100 \text{ GeV})^{12}} \sim 10^{-20},$$

Observed in K^0, B^0 system



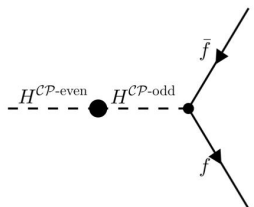
Motivation for searching CPV: The role of the Higgs

'Vanilla' Electroweak Baryogenesis



Reveal CPV: From Theory to experiments

Choose a theoretical model: SMEFT



$$\mathcal{L}_{\text{SMEFT}} = \mathcal{L}_{\text{SM}} + \sum_{d>4} \sum_i \frac{C_i O_i^{(d)}}{\Lambda^{d-4}},$$

Dimension-Six CPV operators

$$O_{\tilde{G}}, O_{\tilde{W}}, O_{\varphi\tilde{G}}, O_{\varphi\tilde{W}}, O_{\varphi\tilde{B}}, O_{\varphi\tilde{W}B}, \hat{O}_{t\varphi}, \hat{O}_{tG}, \hat{O}_{tW}$$

Choose a sensitive channel:

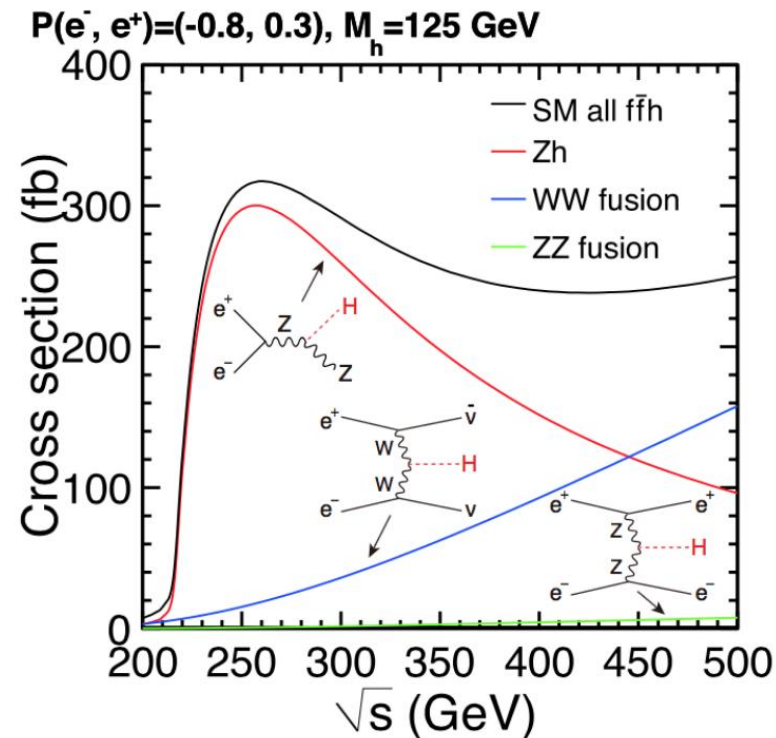
At $\sqrt{s} = 250 \text{ GeV}$ $\rightarrow$ ZH production

$$\begin{aligned}\tilde{\mathcal{O}}_{\Phi\tilde{B}} &= \Phi^\dagger \Phi B^{\mu\nu} \Phi \tilde{B}_{\mu\nu}, \\ \tilde{\mathcal{O}}_{\Phi\tilde{W}} &= \Phi^\dagger \Phi W^{i\mu\nu} \tilde{W}_{\mu\nu}^i, \\ \tilde{\mathcal{O}}_{\Phi\tilde{W}B} &= \Phi^\dagger \sigma^i \tilde{W}^{i\mu\nu} B_{\mu\nu},\end{aligned}$$

After EWSB



$$\begin{aligned}C_{H\tilde{Z}\tilde{Z}} &= c_w^2 C_{H\tilde{W}} + s_w (c_w C_{H\tilde{W}B} + s_w C_{H\tilde{B}}) \\ C_{H\tilde{A}\tilde{A}} &= c_w^2 C_{H\tilde{B}} + s_w (s_w C_{H\tilde{W}} - c_w C_{H\tilde{W}B}) \\ C_{H\tilde{A}\tilde{Z}} &= 2s_w c_w (C_{H\tilde{W}} - C_{H\tilde{B}}) + (s_w^2 - c_w^2) C_{H\tilde{W}B}\end{aligned}$$



Reveal CPV: From Theory to experiments

Observable

$$\mathcal{M} = \mathcal{M}_{\text{even}} + \mathcal{M}_{\text{odd}} \quad \text{CP : } \mathcal{M}_{\text{even}} \rightarrow \mathcal{M}_{\text{even}}^*, \quad \mathcal{M}_{\text{odd}} \rightarrow -\mathcal{M}_{\text{odd}}^*$$

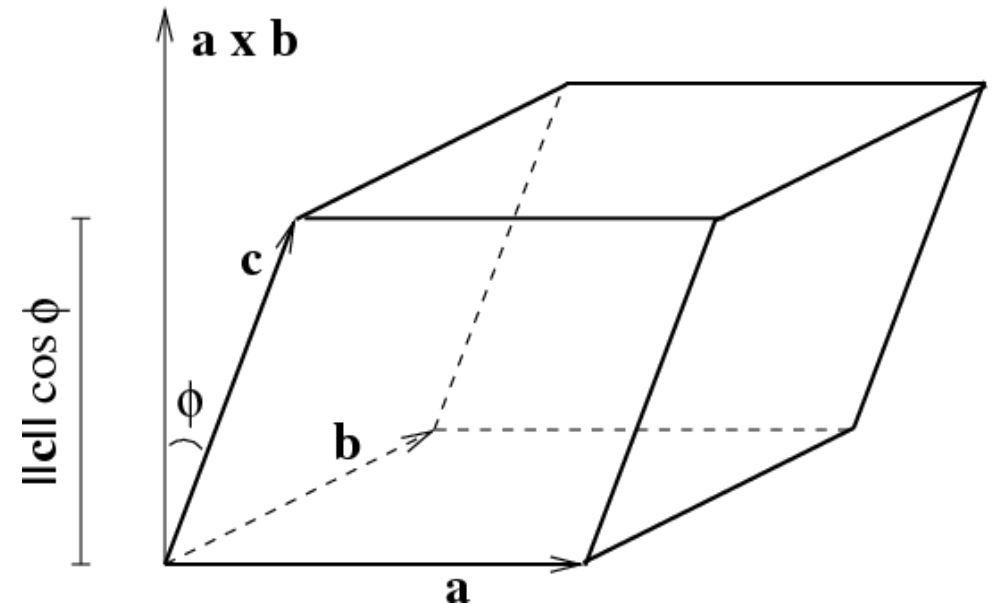
$$|\mathcal{M}|^2 = |\mathcal{M}_{\text{even}}|^2 + |\mathcal{M}_{\text{odd}}|^2 + 2 \operatorname{Re}(\mathcal{M}_{\text{even}} \mathcal{M}_{\text{odd}}^*)$$

$$\Delta_{\text{CP}} \propto \operatorname{Re}(\mathcal{M}_{\text{even}} \mathcal{M}_{\text{odd}}^*)$$

$$\mathcal{M}_{\text{odd}} \sim i C_{\text{CP}} \varepsilon_{\mu\nu\rho\sigma} p_a^\mu p_b^\nu p_c^\rho p_d^\sigma \propto \vec{p}_a \cdot (\vec{p}_b \times \vec{p}_c)$$

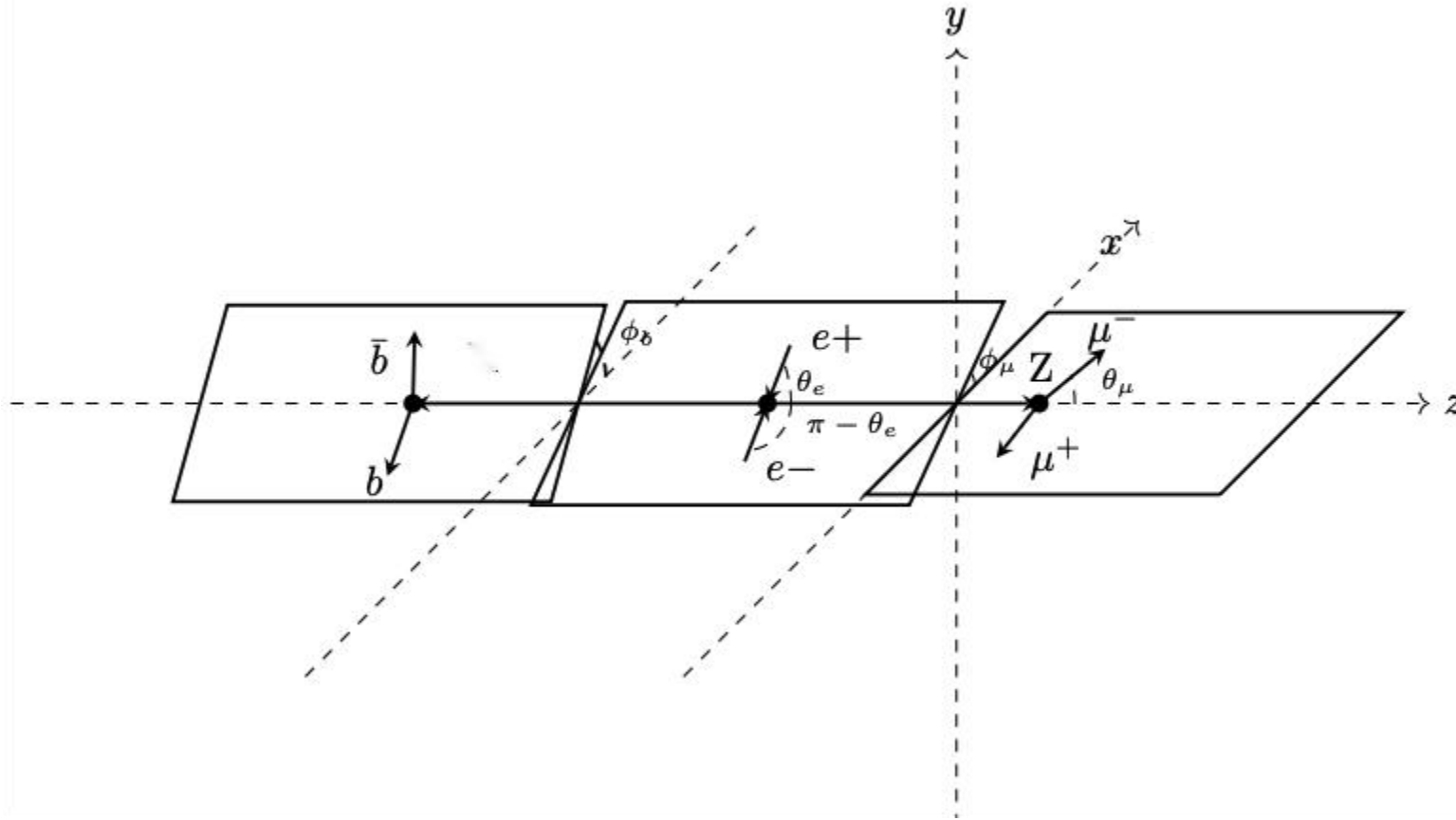
Signed azimuthal angle

$$\Delta\phi_{\ell\ell} = \phi(\ell_1) - \phi(\ell_2) \text{ for } \eta(\ell_1) > \eta(\ell_2),$$



Reveal CPV: From Theory to experiments

Frame Dependant of the angular observable

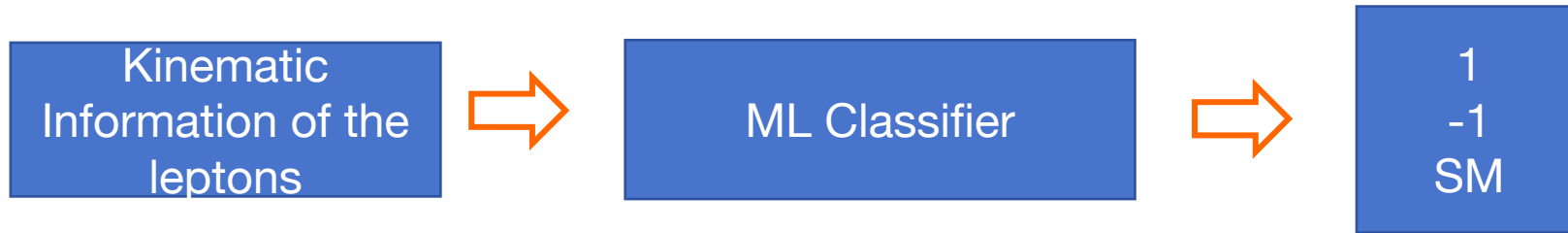


Reveal CPV: From Theory to experiments

Optimal ML-learned Observable

$$|\mathcal{M}|^2 = |\mathcal{M}_{\text{even}}|^2 + |\mathcal{M}_{\text{odd}}|^2 + \boxed{2 \operatorname{Re}(\mathcal{M}_{\text{even}} \mathcal{M}_{\text{odd}}^*)}$$

1
↓ CP
-1



$$O_{\text{ML}} = P_+ - P_- \quad -1 \leq O_{\text{ML}} \leq 1$$

Binary Class: $P_+ + P_- = 1$
Multi Class: $P_+ + P_- + P_{\text{SM}} = 1$

Neural Network VS **Decision Tree**



Reveal CPV: From Theory to experiments

Limit Setting

The profile likelihood ratio

$$\lambda(c) = -2 \log \frac{L(c)}{L_{\max}} \quad L(c) = \prod_{i=1}^n \frac{e^{-\lambda_i}}{n_i!} \lambda_i^{n_i}$$

where n_i is the number of observed events and λ_i is the prediction under a given model with parameter c .

The theoretical number $\lambda_i(c)$ will be evaluated by

$$\lambda_i(c) = \lambda_i^{\text{SM}} + c \cdot \lambda_i^{\text{int}}(c=1) + c^2 \cdot \lambda_i^{\text{EFT}}(c=1)$$

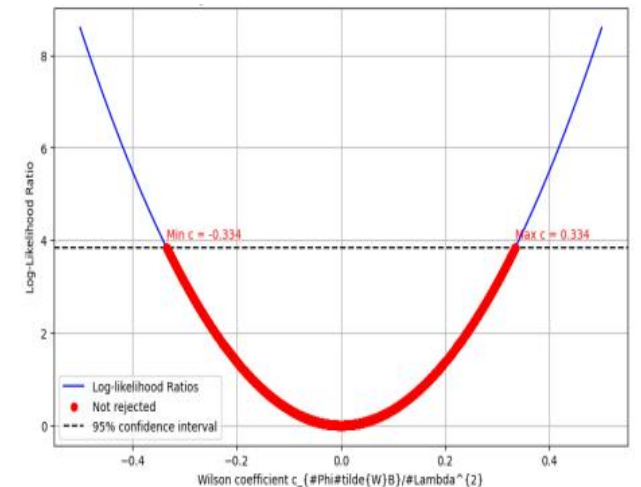
Combine the channels

$$L_{\text{combined}}(c) = L_{\mu\mu}(c) L_{ee}(c)$$

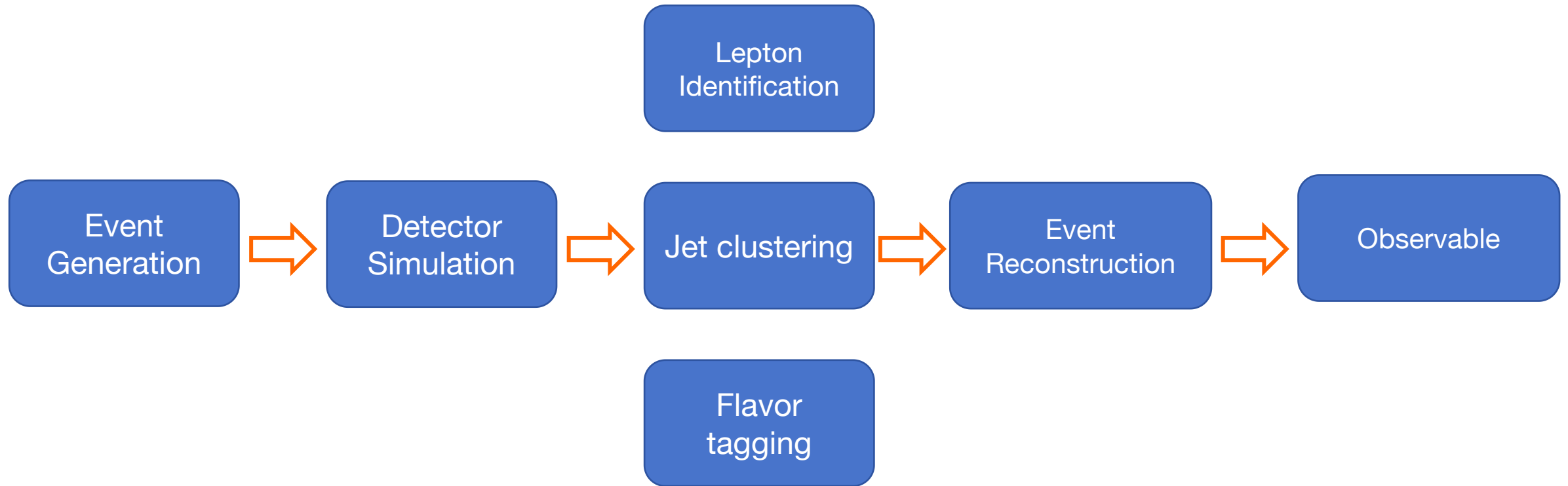
95% Confidence Interval

Following the Wilks' theorem with 1 d.o.f.,

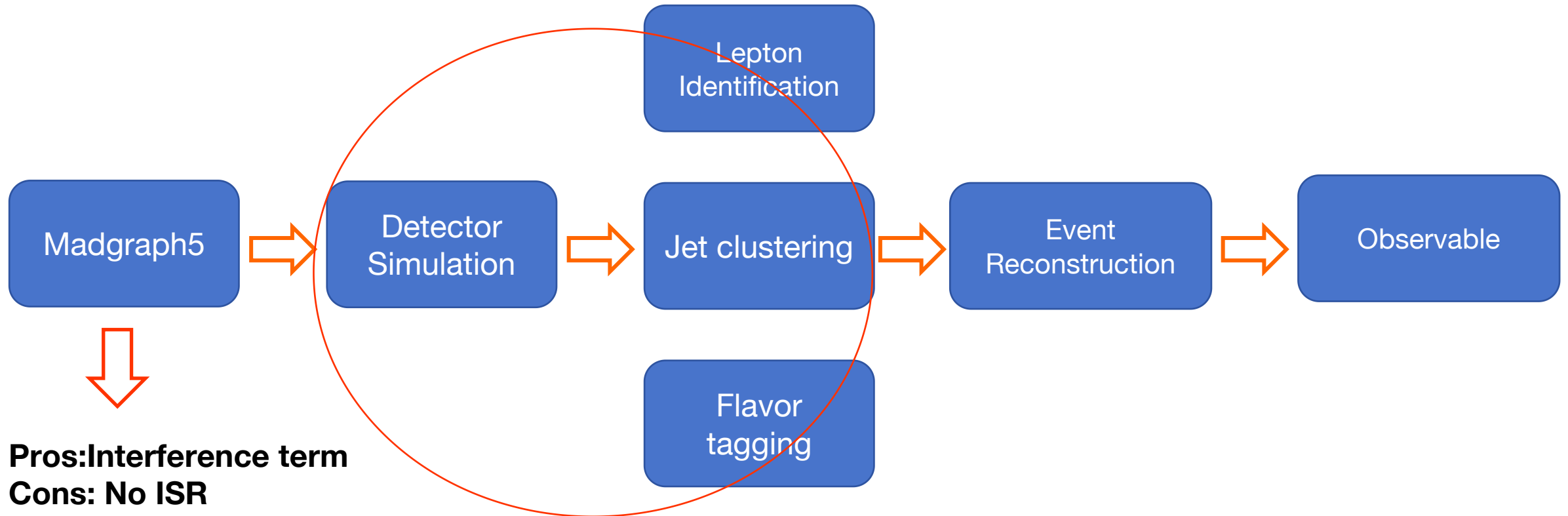
$$\lambda_{\text{all}}(c) = -2 \sum_{i=1}^4 \log \left(\frac{\mathcal{L}_{i,\mu\mu}(c) L_{i,ee}(c)}{\mathcal{L}_{i,\mu\mu,\max} L_{i,ee,\max}} \right) < 3.84$$



Study with Pseudo experiment



Study with Pseudo experiment



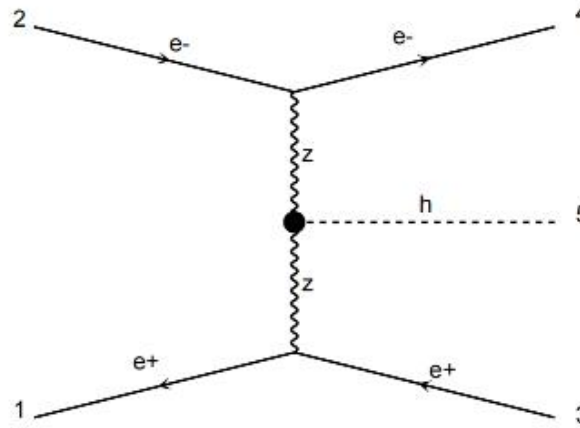
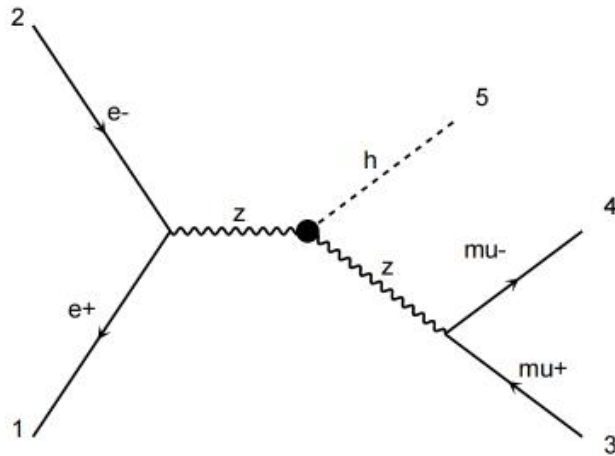
IDEA for FCC
ILCgen for ILC



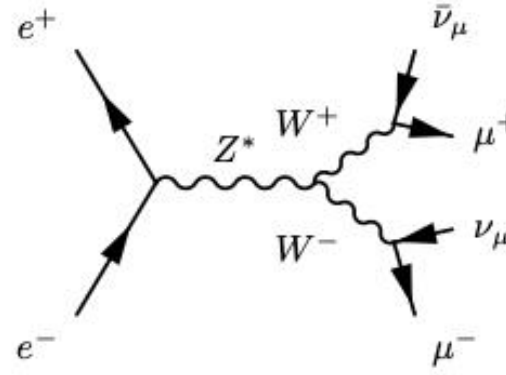
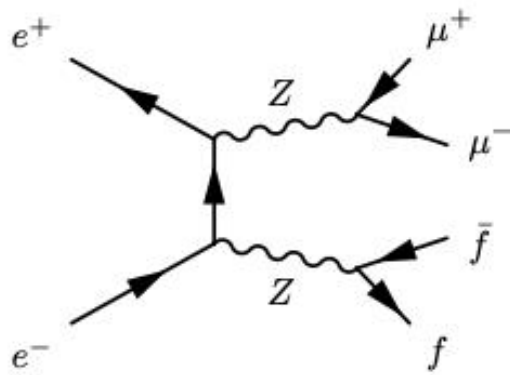
Event Generation: Inclusive

Signal

$e^+ e^- \rightarrow l^+ l^- h$



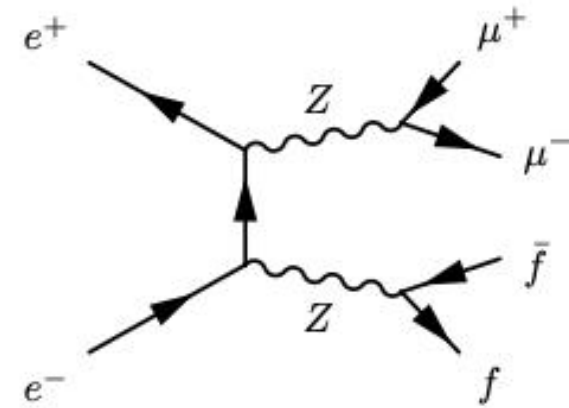
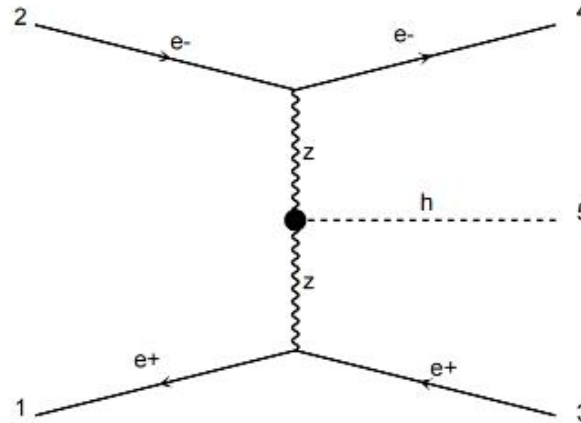
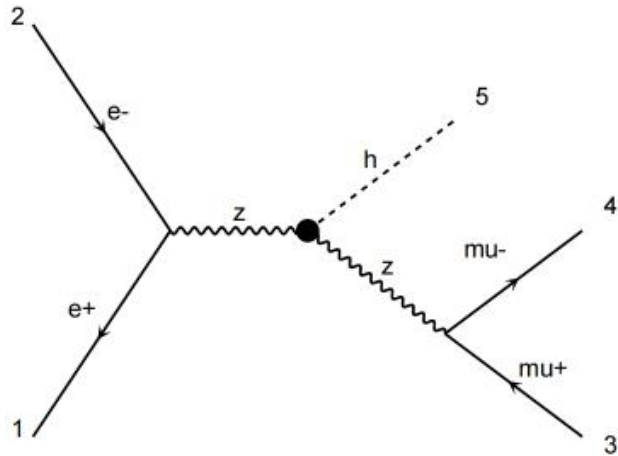
Background



Event Generation: $H \rightarrow b\bar{b}$

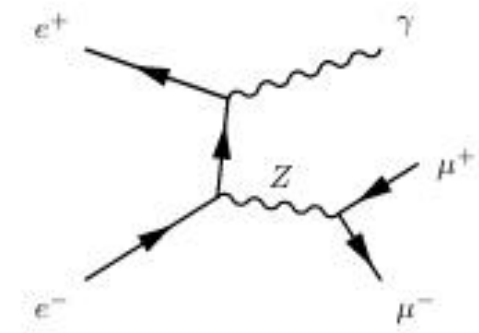
Signal

$$e^+ e^- \rightarrow l^+ l^- b \bar{b}$$



Event Selection

Cut	Purpose
$ M_{\ell\ell} - M_Z < 15 \text{ GeV}$	Select signal events
$ M_{\text{recoil}} - M_H < 5 \text{ GeV}$	
2 b -tagged jets, $p_T^{\text{jet}} > 20 \text{ GeV}$	For $H \rightarrow b\bar{b}$
$p_T^{\ell\ell} > 10 \text{ GeV}$	Reject photon background
$ p_z^{\ell\ell} < 70 \text{ GeV}$	
$\arccos(\vec{n}_{\ell_1} \cdot \vec{n}_{\ell_2}) > 100^\circ$	
$\arccos((\vec{n}_{\ell_1} \times \vec{n}_{\ell_2}) \cdot \hat{k}) > 10^\circ$	



A ISR correction factor :

$$k_{\text{ISR}} = \frac{\sigma_{ZH}^{\text{ISR on}}}{\sigma_{ZH}}$$



Event Selection

SM

Processes	lepton pair	cross section/ fb	*Generated	Reconstructed Pairs	Cut for Signal	Cut for Photon	Efficiency (From reconstruction)	Normalized Yields (FCC)
hz	mu	15.0518	7943	2836	2526	2448	86.3%	40697
	e			2611	2257	2204	84.4%	36572
zz	mu	225.065	118772	36797	200	174	0.47%	2876
	e			57158	44	35	0.06%	591
ww	mu	707.35	373285	81738	912	818	1.00%	13576
	e			80369	849	770	0.96%	12541
total	----	947.4668	500000	261509	6788	6449	2.4%	106856



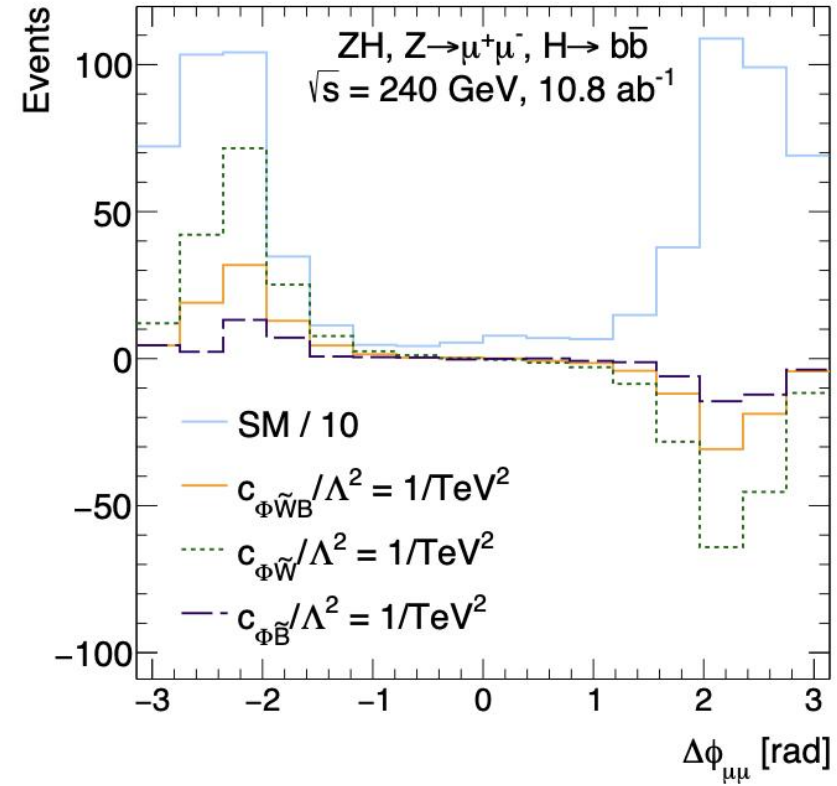
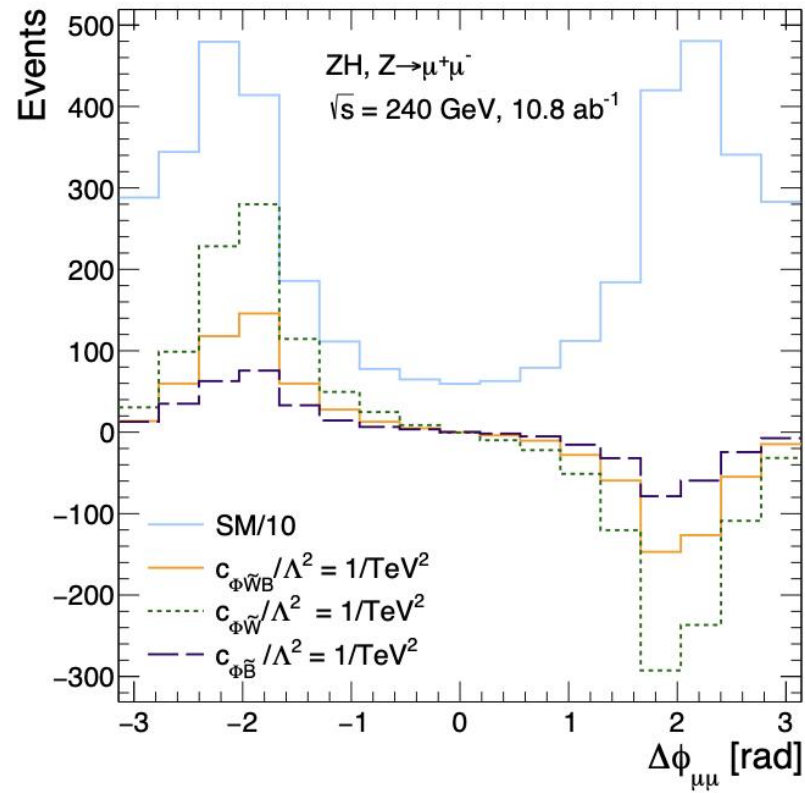
Event Selection

EFT

	lepton pair	Generated	Reconstruct ed	Cut for Signal	Cut for Photon	Efficiency
chwtl=1	mu	187110	138203	121172	116484	84.28%
	e	312890	208277	152364	146681	70.43%
chwbtl=1	mu	114602	85717	71089	67978	79.31%
	e	385398	267526	172653	167167	62.49%
chbtl=1	mu	101054	76082	62458	59922	78.76%
	e	398946	278673	174402	168559	60.49%



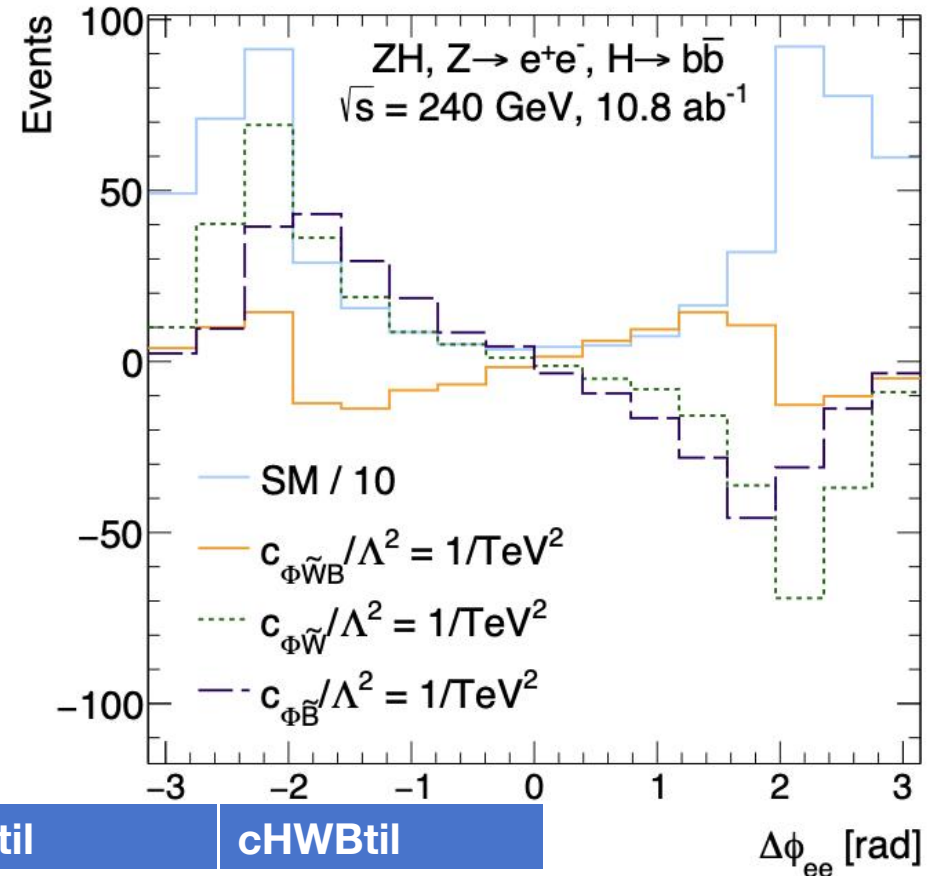
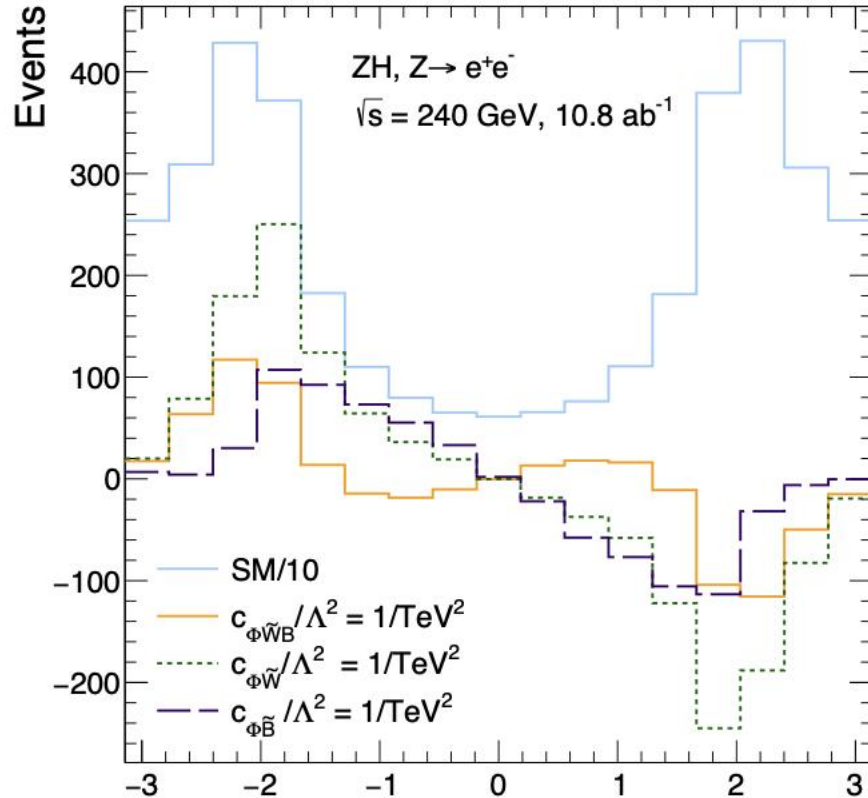
Angular Observable: Comparison of H->inclusive and H->bb



$1/\Lambda^2$	cHWtil	cHBtil	cHWBtil
H->Inclusive	(-0.24,0.24)	(-0.89,0.89)	(-0.47,0.47)
H->bb	(-0.46,0.46)	(-2.2,2.2)	(-1.0,1.0)



Angular Observable: Comparison of H->inclusive and H->bb

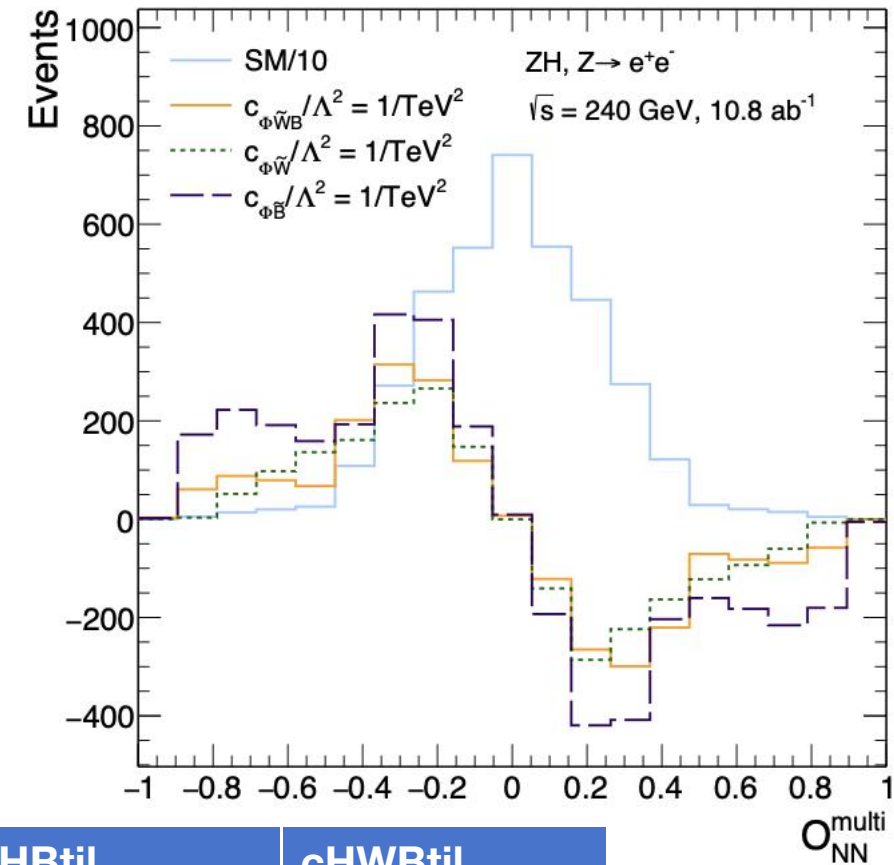
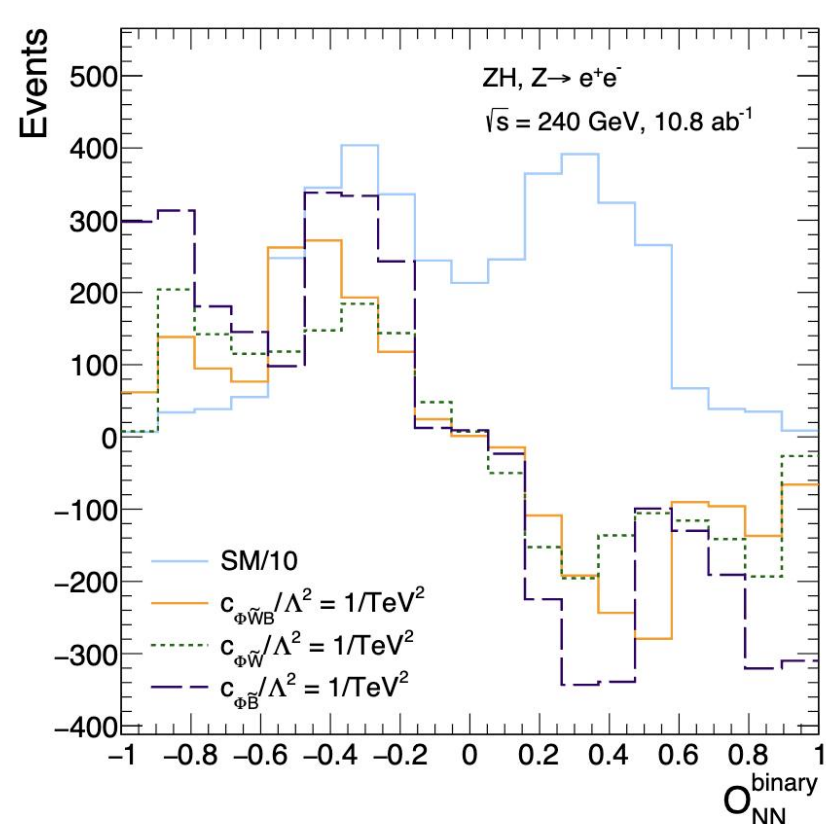


$1/\Lambda^2$	cHWtil	cHBtil	cHWBtil
H->Inclusive	(-0.25,0.25)	(-0.36,0.36)	(-0.57,0.57)
H->bb	(-0.35,0.35)	(-0.31,0.31)	(-0.67,0.67)

$$\Delta_{CP} \propto \text{Re}(\mathcal{M}_{\text{even}} \mathcal{M}_{\text{odd}}^*)$$



ML-learned Observable: Comparision of the binary and multi class

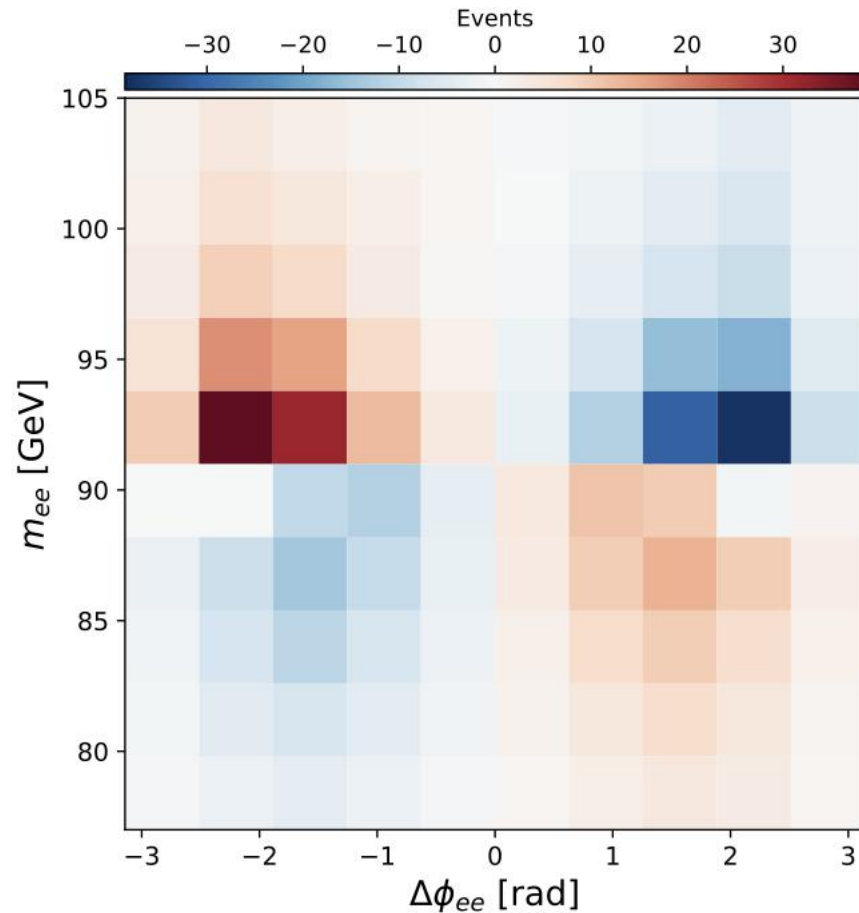


$1/\Lambda^2$	cHWtil	cHBtil	cHWBtil
Binary-Class	(-0.13,0.13)	(-0.08,0.08)	(-0.12,0.12)
Multi-Class	(-0.13,0.13)	(-0.07,0.07)	(-0.12,0.12)
$\Delta\Phi$	(-0.17,0.17)	(-0.33,0.33)	(-0.36,0.36)

Comined
lepton



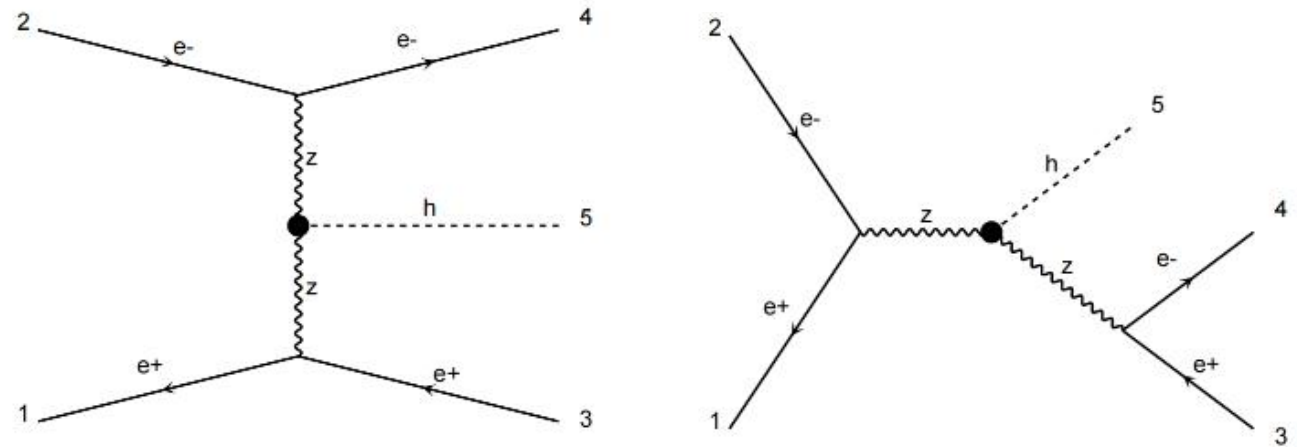
Details on the interference structure



$$C_{H\tilde{Z}\tilde{Z}} = c_w^2 C_{H\tilde{W}} + s_w (c_w C_{H\tilde{W}B} + s_w C_{H\tilde{B}})$$

$$C_{H\tilde{A}\tilde{A}} = c_w^2 C_{H\tilde{B}} + s_w (s_w C_{H\tilde{W}} - c_w C_{H\tilde{W}B})$$

$$C_{H\tilde{A}\tilde{Z}} = 2s_w c_w (C_{H\tilde{W}} - C_{H\tilde{B}}) + (s_w^2 - c_w^2) C_{H\tilde{W}B}$$



Lack of the Luminosity

H->Inclusive

Observable	Luminosity ab^{-1}	chwtl	chwbtl	chbtl
Angular	3	(-0.33, 0.33)	(-0.69, 0.69)	(-0.63, 0.63)
	10.8	(-0.17, 0.17)	(-0.36, 0.36)	(-0.33, 0.33)
ML-binary	3	(-0.25,0.25)	(-0.23,0.23)	(-0.14,0.14)
	10.8	(-0.13,0.13)	(-0.12,0.12)	(-0.08,0.08)
ML-multi	3	(-0.24,0.24)	(-0.22,0.22)	(-0.14,0.14)
	10.8	(-0.13,0.13)	(-0.12,0.12)	(-0.07,0.07)

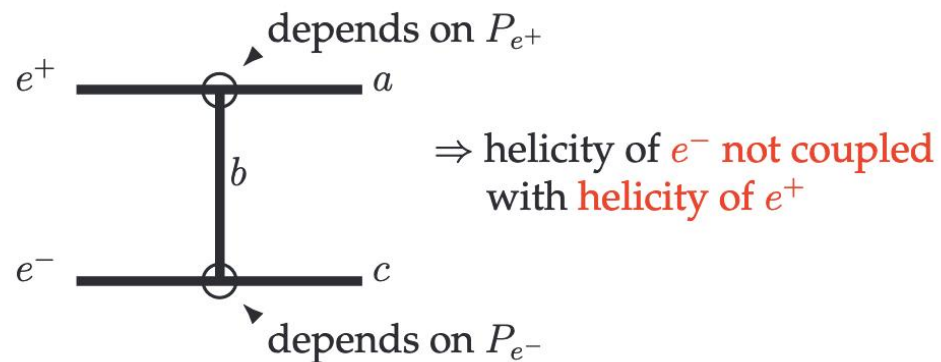
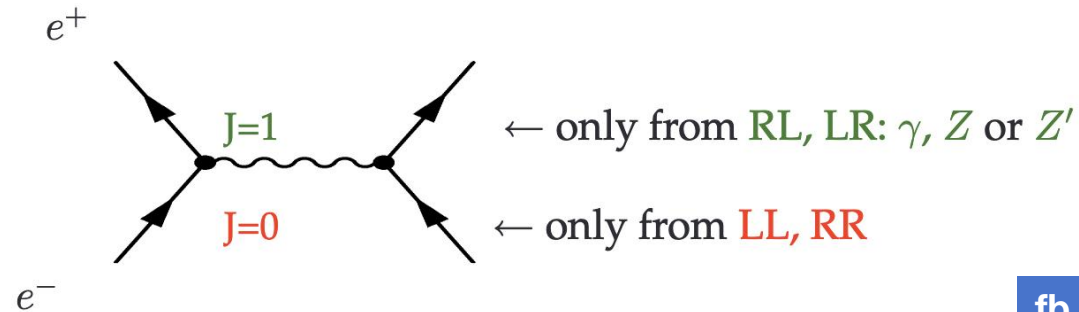
Around a factor of 2 larger than the FCC-ee's number



Take Out Polarised Beam! (Longitude)

General Effect on the total cross-section

Helicity Selection



Left-Right asymmetry

$$g_Z^L \simeq -0.538$$

$$g_Z^R \simeq 0.462$$

fb	non	(-0.8,0.3)	(+0.8,-0.3)	(-0.8,-0.3)	(+0.8,+0.3)
hz	15.0518	20.6699	16.3393	12.5851	10.6149
zz	225.0647	308.7441	216.6222	208.7676	166.9780
ww	707.3500	1640.643	114.5366	88.3936	190.8008



Take Out Polarised Beam!

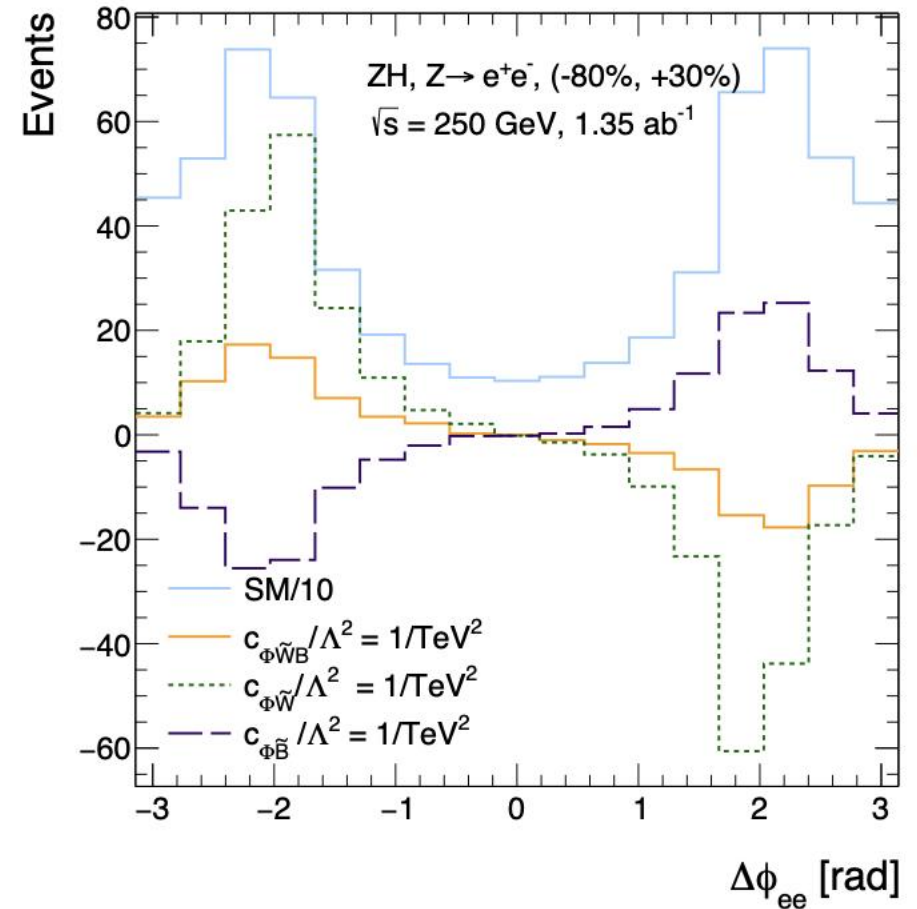
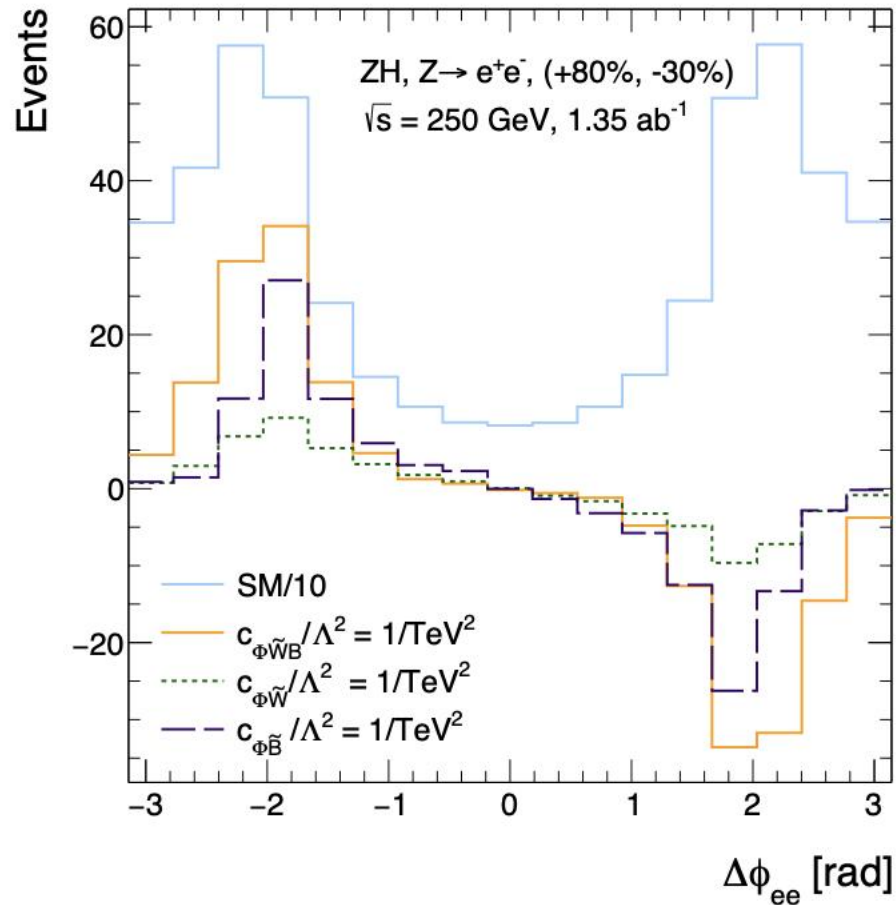
Effects on the EFTs

$$\begin{aligned}C_{H\tilde{Z}\tilde{Z}} &= c_w^2 C_{H\tilde{W}} + s_w (c_w C_{H\tilde{W}B} + s_w C_{H\tilde{B}}) \\C_{H\tilde{A}\tilde{A}} &= c_w^2 C_{H\tilde{B}} + s_w (s_w C_{H\tilde{W}} - c_w C_{H\tilde{W}B}) \\C_{H\tilde{A}\tilde{Z}} &= 2s_w c_w (C_{H\tilde{W}} - C_{H\tilde{B}}) + (s_w^2 - c_w^2) C_{H\tilde{W}B}\end{aligned}$$

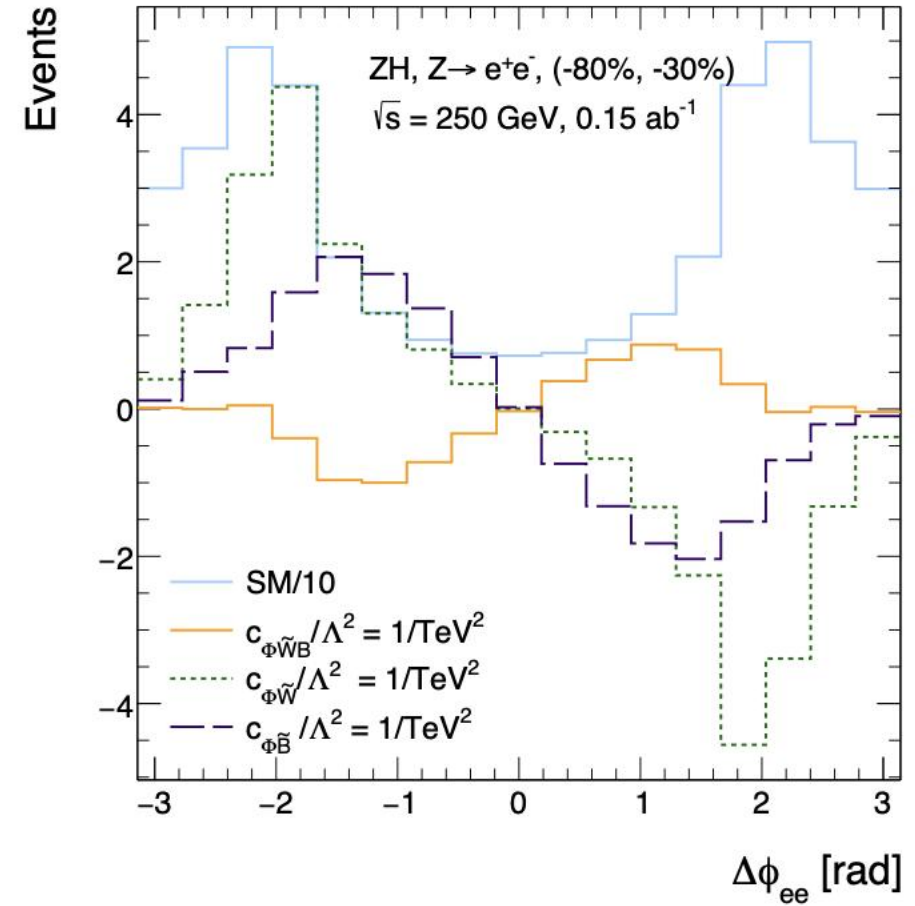
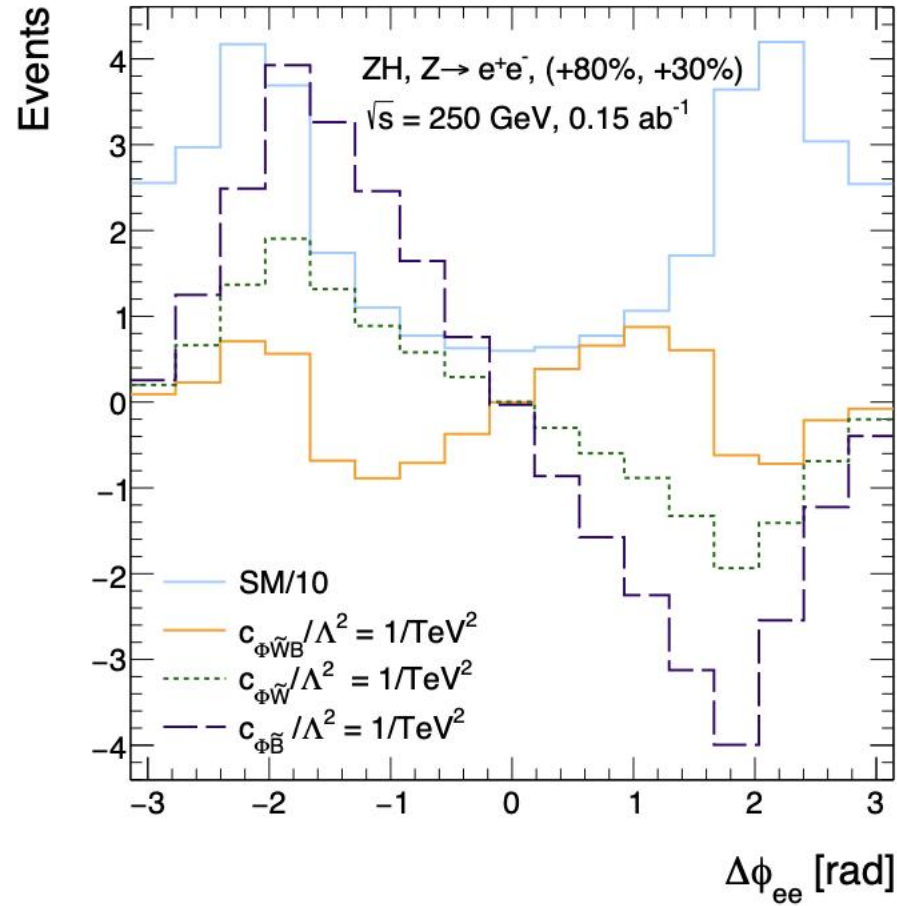
- $\mathcal{O}_{\phi\tilde{W}}$ will benefit from the left-right asymmetry when the ZH production is enhanced, while $\mathcal{O}_{\phi\tilde{W}B}$ will benefit from the ZH production is suppressed.
- The ZH production and the VBF for $\mathcal{O}_{\phi\tilde{W}B}$ will cancel each other, but it will experience a sign-flip if when the dominant process changed to each other
- $\mathcal{O}_{\phi\tilde{B}}$ mainly contributes to the VBF channel which is not affected by the helicity selection rules



Angular observable with Polarised Beam



Angular observable with Polarised Beam



ML-learned Observable: The precision

Binary Class

	Lepton	non	(-0.8,0.3)	(0.8,-0.3)	(-0.8,-0.3)	(0.8,0.3)
$c_{HWtil}=1$	mu	0.67	0.68	0.68	0.68	0.68
	e	0.68	0.72	0.69	0.67	0.69
$c_{HWBtil}=1$	mu	0.68	0.60	0.68	0.64	0.68
	e	0.69	0.82	0.67	0.69	0.64
$c_{HBtil}=1$	mu	0.57	0.64	0.68	0.58	0.65
	e	0.69	0.75	0.74	0.66	0.67

Multi Class

	Lepton	non		(-0.8,0.3)		(0.8,-0.3)		(-0.8,-0.3)		(0.8,0.3)	
		EFT	SM	EFT	SM	EFT	SM	EFT	SM	EFT	SM
$c_{HWtil}=1$	mu	0.48	0.58	0.50	0.57	0.47	0.56	0.48	0.57	0.49	0.55
	e	0.51	0.52	0.54	0.53	0.54	0.50	0.51	0.53	0.53	0.51
$c_{HWBtil}=1$	mu	0.49	0.54	0.44	0.48	0.49	0.56	0.46	0.52	0.51	0.54
	e	0.55	0.52	0.67	0.61	0.50	0.51	0.57	0.59	0.48	0.48
$c_{HBtil}=1$	mu	0.41	0.52	0.48	0.54	0.48	0.56	0.41	0.51	0.46	0.59
	e	0.59	0.51	0.63	0.50	0.59	0.52	0.56	0.53	0.53	0.51



Take Out Polarised Beam!

Final Limits

Observable	Luminosity	chwtl	chwbtl	chbtl
Angular	3 ab	(-0.33, 0.33)	(-0.69, 0.69)	(-0.63, 0.63)
	10.8 ab	(-0.17, 0.17)	(-0.36, 0.36)	(-0.33, 0.33)
	LCF	(-0.28, 0.28)	(-0.38, 0.38)	(-0.35, 0.35)
ML-binary	3 ab	(-0.25,0.25)	(-0.23,0.23)	(-0.14,0.14)
	10.8 ab	(-0.13,0.13)	(-0.12,0.12)	(-0.08,0.08)
	LCF	(-0.21,0.21)	(-0.19,0.19)	(-0.11,0.11)
ML-multi	3 ab	(-0.24,0.24)	(-0.22,0.22)	(-0.14,0.14)
	10.8 ab	(-0.13,0.13)	(-0.12,0.12)	(-0.07,0.07)
	LCF	(-0.20,0.20)	(-0.18,0.18)	(-0.10,0.10)



Play Cards : Luminosity vs Polarization

In early this year, when FCC planned $L=5\text{ab}^{-1}$, we get...(H->bb)

Observable	Luminosity	cHWtil	cHWBtil	cHBtil
Rapidity-ordered azimuthal leptons	2 ab^{-1} non-polarised	(-0.62, 0.62)	(-1.45, 1.45)	(-1.11, 1.11)
	5 ab^{-1} non-polarised	(-0.40, 0.40)	(-0.92, 0.92)	(-0.70, 0.70)
	2 ab^{-1} ILC combined	(-0.44, 0.44)	(-1.11, 1.11)	(-0.85, 0.85)
CatBoost Binary	2 ab^{-1} non-polarised	(-0.41, 0.41)	(-0.35, 0.35)	(-0.22, 0.22)
	5 ab^{-1} non-polarised	(-0.26, 0.26)	(-0.22, 0.22)	(-0.14, 0.14)
	2 ab^{-1} ILC combined	(-0.25, 0.25)	(-0.32, 0.32)	(-0.22, 0.22)
CatBoost Multi	2 ab^{-1} non-polarised	(-0.35, 0.35)	(-0.31, 0.31)	(-0.18, 0.18)
	5 ab^{-1} non-polarised	(-0.22, 0.22)	(-0.20, 0.20)	(-0.12, 0.12)
	2 ab^{-1} ILC combined	(-0.22, 0.22)	(-0.26, 0.26)	(-0.18, 0.18)

In middle this year, I was informed FCC planned $L=10.8\text{ab}^{-1}$, we get...(Inclusive)

Observable	Luminosity	chwtl
Angular	10.8 ab	(-0.17, 0.17)
	LCF	(-0.28, 0.28)
ML-binary	10.8 ab	(-0.13,0.13)
	LCF	(-0.21,0.21)
ML-multi	10.8 ab	(-0.13,0.13)
	LCF	(-0.20,0.20)

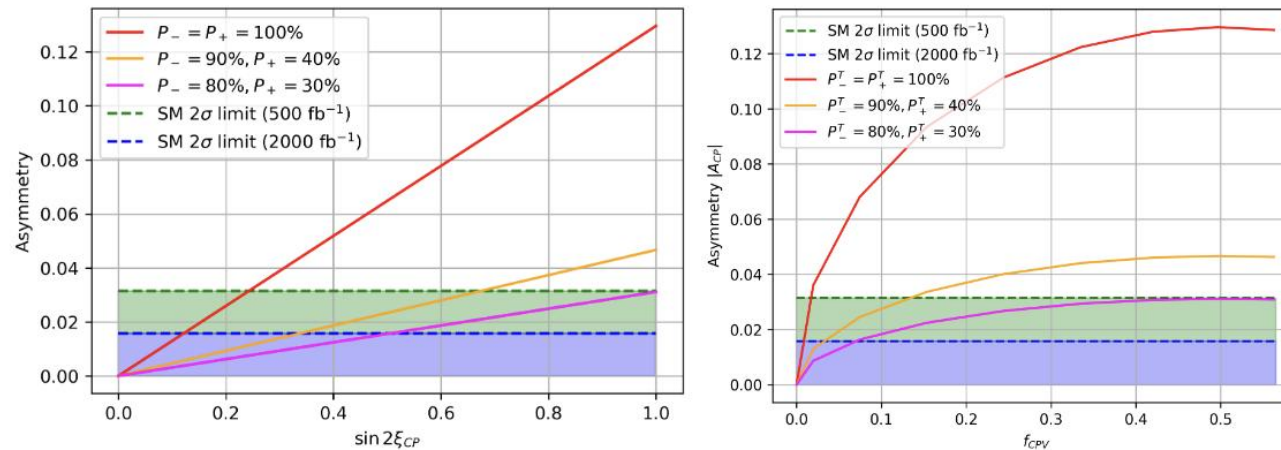


Play Cards : Luminosity vs Polarization

Later, LCF plan to introduce transverse polarization...

$$C_{\text{CP-mix}}^{mn} \propto \epsilon_{\mu\nu\rho\sigma} [(p_{e^-} + p_{e^+})^\mu p_H^\nu p_{\mu^-}^\rho s_{e^-}^\sigma] \propto (\vec{p}_H \times \vec{p}_{\mu^-}) \cdot \vec{s}_{e^-} \propto (\vec{p}_{\mu^+} \times \vec{p}_{\mu^-}) \cdot \vec{s}_{e^-}.$$

$$O_{CP} = \eta_H \sin 2\phi_{\mu^-}.$$



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Need Further detector-level studies with transverse-polarised beam



Thank you for attention!



University
of Glasgow

